

PAPER_1320 — Charged Lepton Flavor Violation BR($\mu \rightarrow e\gamma$)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** C — Particle Physics Open (CLOSED) **Date:** June 11, 2026 **Calculator surface:** `calculate_paradox({"paradox": "clfv"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1320})`

Master Expression

$$\text{BR}(\mu \rightarrow e\gamma) = \Lambda^6 \times \Phi_{\text{res}} = 1.27 \times 10^{-13}$$

Match: just below MEG bound 4.2×10^{-13}

Interpretation

cLFV emerges at the $\Lambda^6 \times \Phi_{\text{res}}$ scale — testable at MEG-II (expected sensitivity 6×10^{-14}).

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$
- $\Lambda_{\text{QCD}} = 0.217 \text{ GeV}$, $v_{\text{Higgs}} = 246 \text{ GeV}$ (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 11, 2026, Youngstown, OH. Subject matter: UQFF derivation of Charged Lepton Flavor Violation BR($\mu \rightarrow e\gamma$).